SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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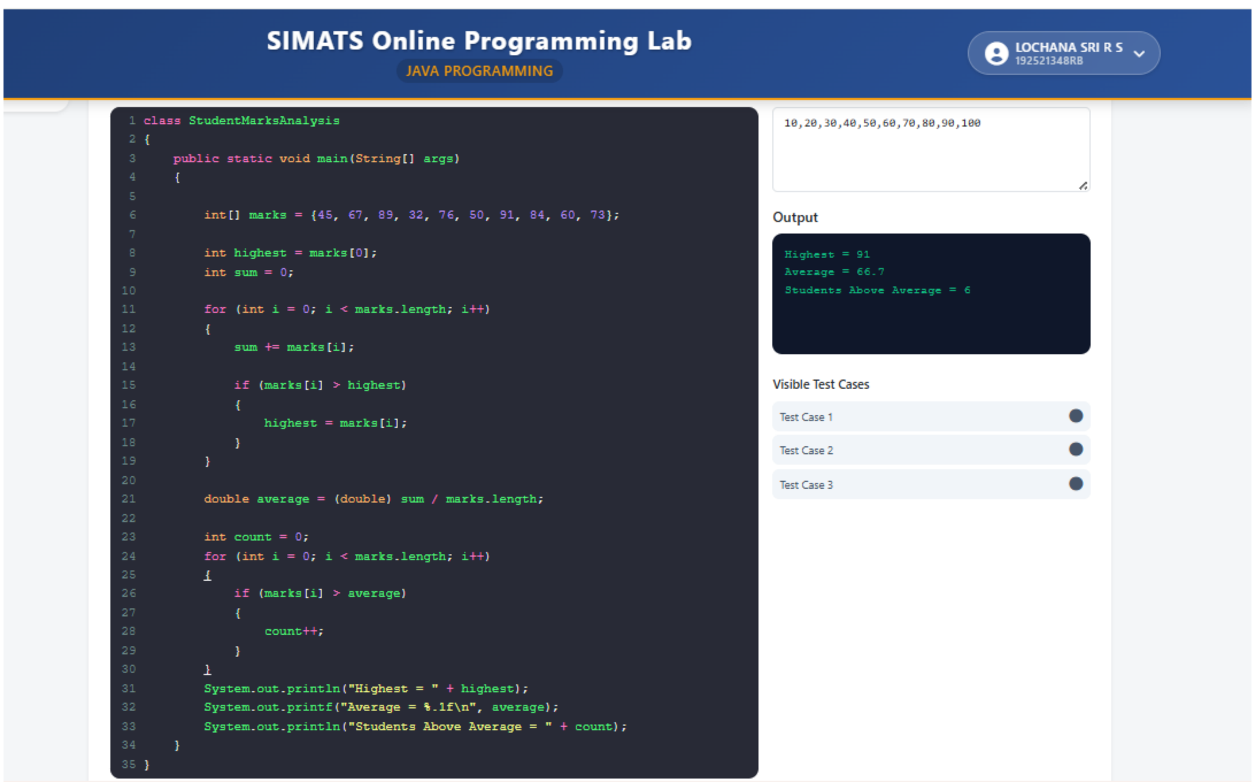
COURSE: CSA0915 -JAVA PROGRAMMING

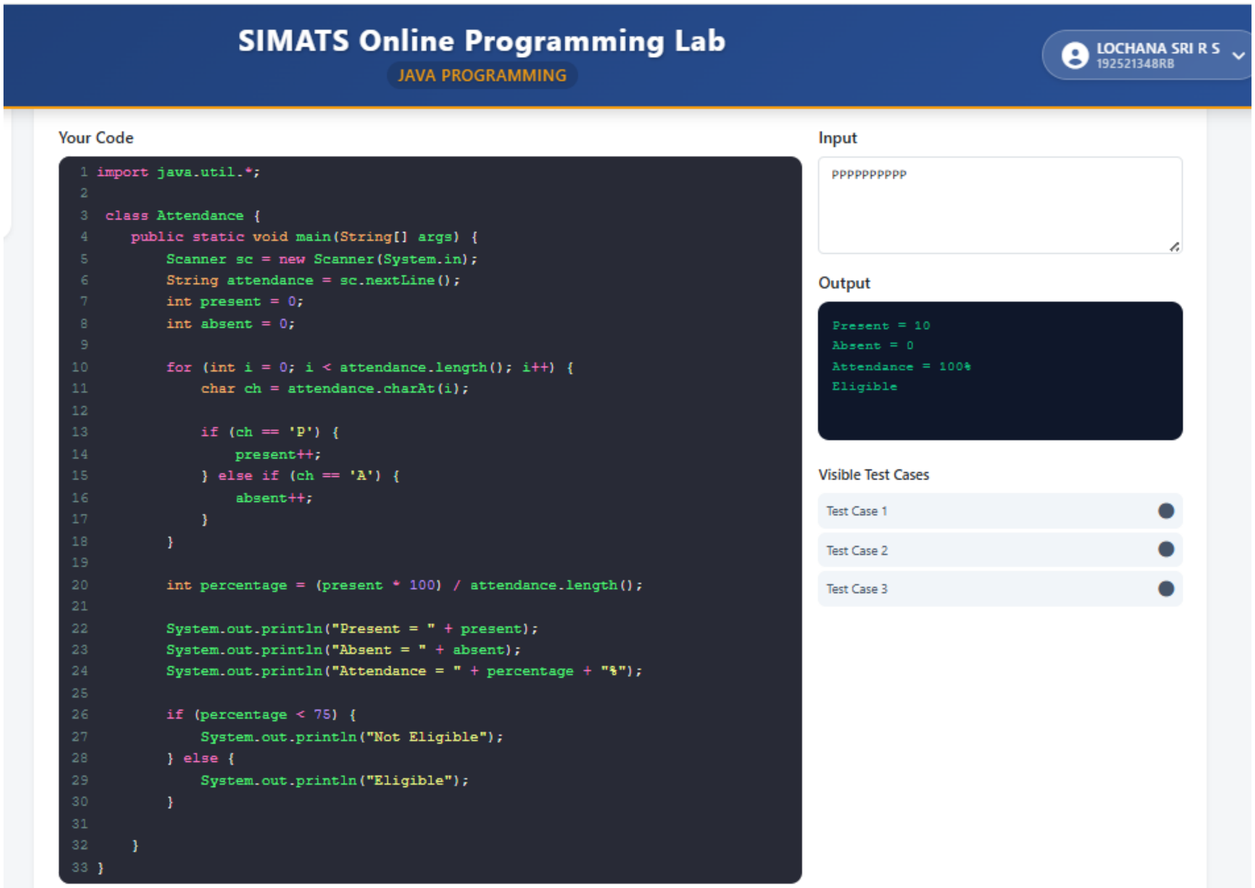
SLOT: B

SCENARIO BASED QUESTIONS

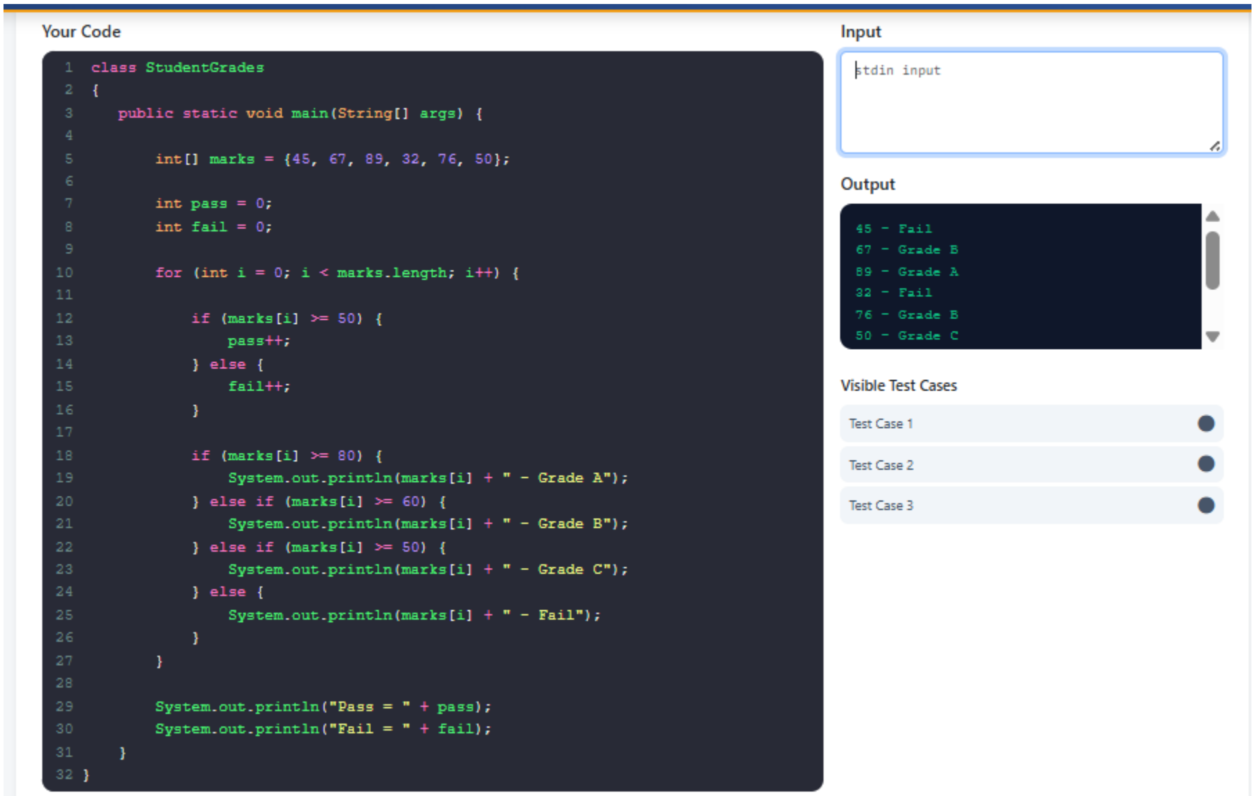
1. A teacher has stored the marks of 10 students in an array. She wants to analyze the performance of the class by finding the highest mark, calculating the average mark, and counting how many students scored above the average.



2. A school records attendance using a string such as 'PPAAPPAPPP'. Count Present, Absent, attendance percentage and display 'Not Eligible' if below 75%.

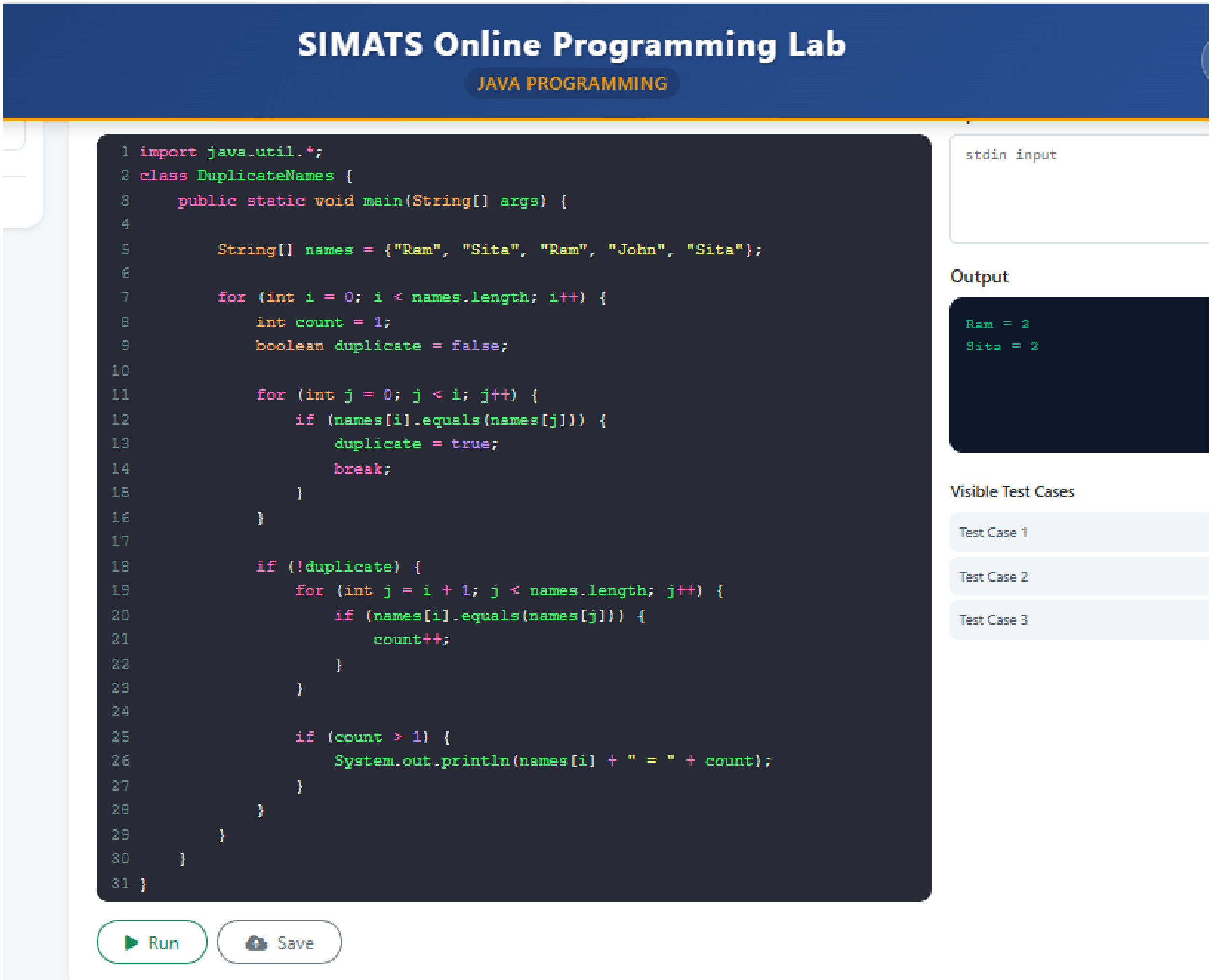


3. A teacher stores student marks in an array. Count pass/fail and assign grades (A,B,C,Fail).




```
class StudentGrades
{
    public static void main(String[] args) {

        int[] marks = {45, 67, 89, 32, 76, 50};

        int pass = 0;
        int fail = 0;

        for (int i = 0; i < marks.length; i++) {

            if (marks[i] >= 50) {
                pass++;
            } else {
                fail++;
            }

            if (marks[i] >= 80) {
                System.out.println(marks[i] + " - Grade A");
            } else if (marks[i] >= 60) {
                System.out.println(marks[i] + " - Grade B");
            } else if (marks[i] >= 50) {
                System.out.println(marks[i] + " - Grade C");
            } else {
                System.out.println(marks[i] + " - Fail");
            }
        }

        System.out.println("Pass = " + pass);
        System.out.println("Fail = " + fail);
    }
}
```

4. An organization maintains employee names in an array. Identify duplicate names and count occurrences.



5. A bank stores transactions where positive values are deposits and negative values are withdrawals. Calculate final balance and count deposits and withdrawals.

SIMATS Online Programming Lab

JAVA PROGRAMMING

```java
import java.util.*;
 class BankTransactions {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of transactions: ");
        int n = sc.nextInt();

        int[] transactions = new int[n];

        System.out.println("Enter the transactions:");

        for (int i = 0; i < n; i++) {
            transactions[i] = sc.nextInt();
        }

        int balance = 0;
        int deposits = 0;
        int withdrawals = 0;

        for (int i = 0; i < n; i++) {

            balance = balance + transactions[i];

            if (transactions[i] > 0) {
                deposits++;
            } else if (transactions[i] < 0) {
                withdrawals++;
            }
        }

        System.out.println("Balance = " + balance);
        System.out.println("Deposits = " + deposits);
        System.out.println("Withdrawals = " + withdrawals);
    }
}
```